

3.6.2 Bauddha School of Philosophy

The Bauddha (or Buddhist) school of philosophy is largely based on the teachings of Gautama Buddha. Buddha was born as Siddhārtha during the 4th–5th Century BCE. Although Buddhism originated in ancient India, it later spread to several parts of Asia. There are two forms of Buddhism: the northern form and the southern form. The form of Buddhism prevailing in Nepal, Tibet, China, Korea, Vietnam, Taiwan, Singapore, and Japan is called northern Buddhism (also synonymous with Mahāyāna) while the form prevailing in Sri Lanka and other parts of Southeast Asia including Cambodia, Laos, Myanmar, and Thailand is called Southern Buddhism (also known as Theravāda). The earliest form is the northern version, and it includes several sub-traditions such as Zen, Nichiren, and Shingon. Kaniśka is supposed to have convened a great council of the Northern Buddhists in the 1st century CE. Lalitavistara is an important work composed by the Northern Buddhists sometime during the 2nd–4th century CE. Tibetan Buddhism drifted away from the primitive Buddhism in India and is supposed to have adopted forms and ceremonies, which were unknown to Gautama and his followers.

According to early texts, Gautama was moved by the suffering of life and death. Further, on account of rebirth, this suffering is experienced in an endless cycle of birth–death. His enlightenment showed him the path for liberation from this suffering forever, by reaching a state of Nirvāṇa. Gautama's teachings were initially oral and in the later period, they developed into a complete philosophical system with several treatises written by the followers of the school.

Buddha's philosophy focuses on the means of ending the suffering of the individuals. It is based on four noble truths (catvāri-ārya-satyāni). Figure 3.6 graphically illustrates this. These are elaborated as follows:

1. **There is suffering:** According to Gautama, "Birth is suffering, decay is suffering, illness is suffering, death is suffering. The presence of objects we hate is suffering, not able

to obtain what we desire is suffering." The Buddhist philosophy argues that the human being is a compound of five aggregates and clinging to them leads to suffering. The five aggregates include the following:

- (a) The form made of four elements (earth, water, fire, and air), five sense organs, five attributes of matter (smell, form, sound, taste, and touch), two distinctions of sex (male, female), three essential conditions (thought, vitality, and space) and two means of communication (gesture and speech)
- (b) Consciousness
- (c) Feeling: Sensations of pleasure and pain
- (d) Formation
- (e) Perception and Potentialities which lead to good or bad results

These five aggregates include all physical and mental elements and powers of man and are impermanent in nature. Consciousness arises from other aggregates and mental factors from the contact of consciousness and other aggregates.

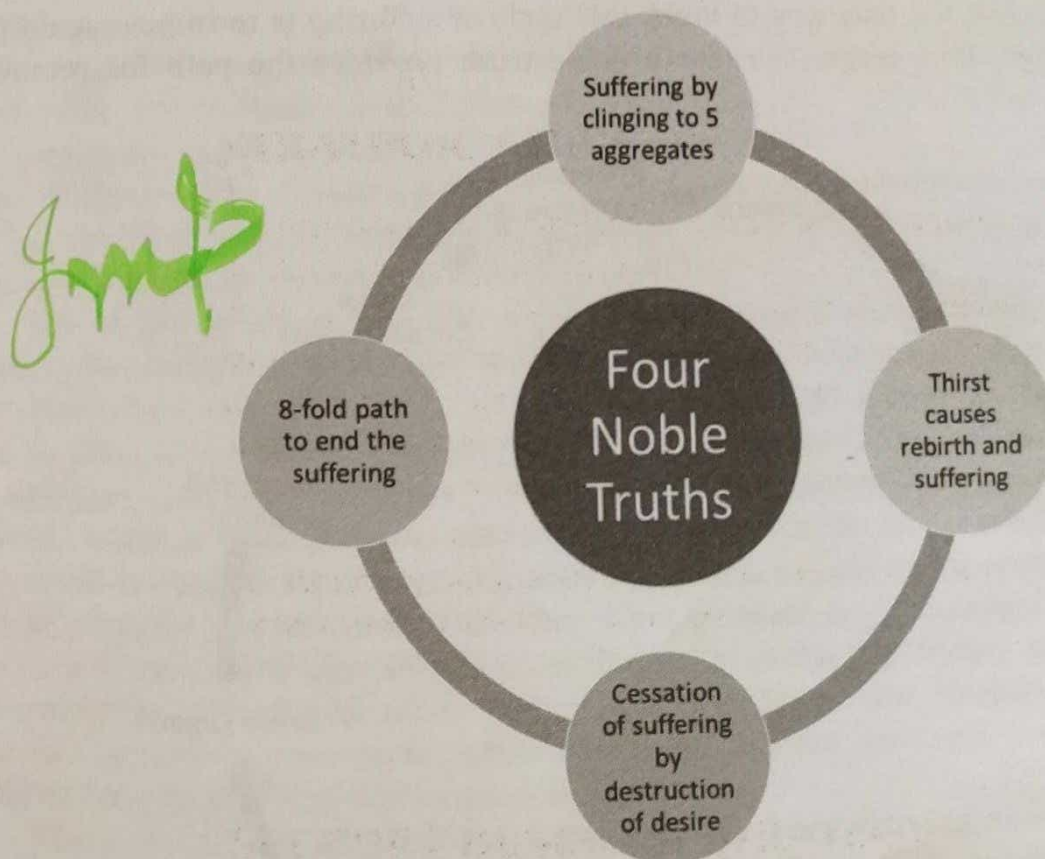


FIGURE 3.6 The Four Noble Truths of Buddhism

2. *There is the cause of suffering:* In Gautama's words, "Thirst leads to rebirth accompanied by pleasure and lust, thirst for pleasure, thirst for existence, thirst for prosperity".
3. *The cessation of suffering:* The cessation of suffering will be possible with the complete cessation of thirst, which amounts to the absence of passion and complete destruction of desire.

4. *There is a path to end the suffering:* Buddhist philosophy prescribes a holy eight-fold path that enables one to lead a holy moral life and that will lead one to the final goal of liberation. The eightfold path includes right views, right resolve, right speech, right conduct, right livelihood, right effort, right mindfulness, and right concentration.

The ultimate goal in Buddhist philosophy is to reach Nirvāṇa. Nirvāṇa is not a state reached after death, but something that is attainable in this very life. It is the sinless calm state of mind attained due to freedom from desires and passions, a state of perfect peace, goodness, and wisdom. Once a person reaches Nirvāṇa, the cycle of birth and death ends, which is the final goal leading to liberation.

As we have seen above, the central issue in Buddhist philosophy is to strive for the cessation of suffering. The Buddhist philosophy systematically argues how sufferings happen using a cause-effect cycle. Figure 3.7 illustrates this cycle leading to suffering. As seen in the figure, the root cause of the suffering is ignorance. Due to ignorance, the impressions of the previous birth lead to initial consciousness. The body and the mind and the sense organs evolve out of this consciousness. Once the sense organs are in contact with the senses and gather the experiences of life, the thirst for enjoyment drives the process leading to rebirth and suffering. Therefore, the only way to break this cycle of suffering is to remove ignorance by acquiring the right knowledge. The fourth noble truth provides the path for removing ignorance.

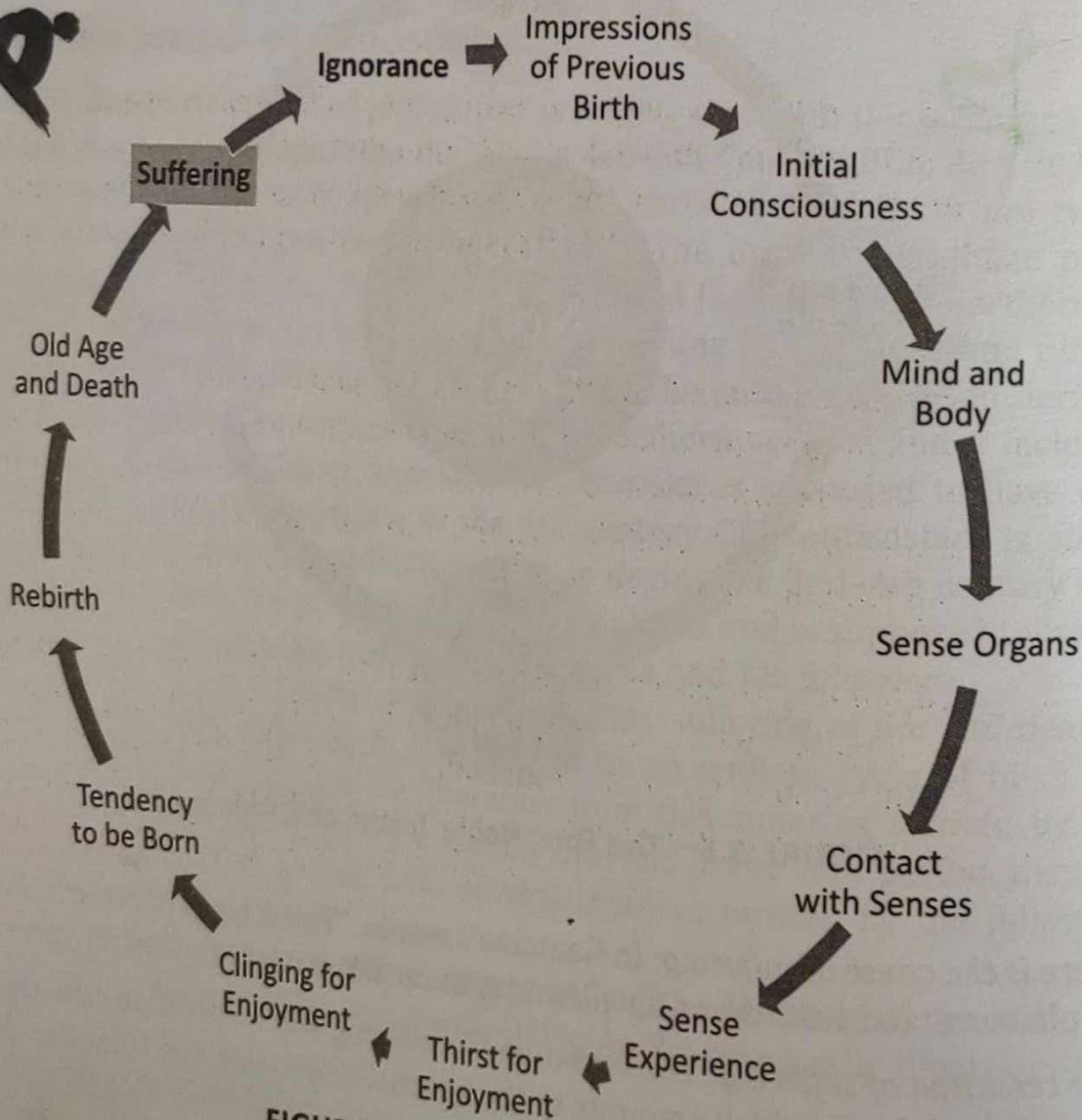


FIGURE 3.7 The Cycle of Suffering

3.6.3 Cārvāka School

Cārvāka school of thought closely maps to the trait of materialism, which emphasises a life of enjoyment based on certain principles and assumptions about life. The word Cārvāka literally means 'sweet-tongued' (cāru-vāka), in a way signifying that the ideas appear attractive at the outset. This is because the system only advocates two of the four puruṣārthas, pleasure and wealth as objectives of living. This school is also called Lokāyata. The Cārvāka school considers matter as the ultimate reality and rejects the idea that there is a divine or a transcendental power behind the matter, called Prakṛti conceptualised by the Sāṃkhya-Yoga school.

Philosophical systems in India had systematic methods for the presentation of the key concepts in the system. This began with a book (of aphorisms), followed by a growing literature of a few commentaries and sub-commentaries. In the case of the Cārvāka system, we do not seem to have such extensive literature on the school. No text of the Cārvāka school is available to us today which discusses its tenets in totality. The absence of canonical texts and a lineage of followers who were able to establish the tenets of the system by constantly engaging in intellectual debates to establish their tenets were perhaps responsible for its decay. Current discussions on this school of thought are based on the scanty fragments available for some analysis⁹. The available material is from texts such as Sarva-darśana-saṅgraha where, during the discussion of these schools, some verses are quoted. Mostly we get to know about this system through refutations from its opponents.

- ♦ No text of the Cārvāka school is available to us today which discusses its tenets in totality.
- ♦ According to Cārvāka school whatever is directly perceivable can only be accepted as valid means of knowledge.

One of the major differences of the Cārvāka school with that of the Vedic schools of philosophy pertains to what is the accepted means of valid knowledge. Unlike all other schools, Cārvāka school considers only direct perception through senses as pramāṇa¹⁰. Essentially this implies that whatever is directly perceivable can only be accepted as valid means of knowledge. This has significant implications for metaphysics. On account of this, Cārvāka school considers matter as the only reality using which the world is made of. Furthermore, the world is constituted of only four basic categories, namely, earth, water, fire, and air, which are all physical and directly perceivable. Ether or space is not accepted as the fifth element because it is not perceivable. Other entities such as the sky, ātman, mind, īśvara, dharma, reincarnation, svarga, and mokṣa that the other schools have accepted are rejected in the Cārvāka system. In essence, all transcendental entities are dismissed using the argument that only direct perception provides valid knowledge.

There is nothing called ātman other than the body. Cārvāka considers the four basic elements of the world as the basic constituents of the body too. When the individual constituents exist in a disjointed state, they are bereft of life and consciousness. However, when these come together the body is formed, and by a peculiar combination of these constituents, the life-breath and consciousness appear in the body. The concept of rebirth and other world is completely dismissed in this system. The arguments primarily stem from the limitation of using only direct perception as the means of valid knowledge. There are no means available for determining the existence of the 'other world'. Moreover,

there is no 'other world' because of the absence of any 'otherworldly' being. Since the existence of consciousness in the other world cannot be substantially established through direct perception, which is the only means available for valid knowledge, these ideas are dismissed.

Once a person dies, there is no afterlife. Pleasure and pain are felt in no other place than the body, hence there is no need to accept something called the ātman which is not confirmed by the pramāṇa. Since there is no ātman and there is no rebirth the highest goal of human life is to have pleasure in one's lifetime. Once the body is confined to the flames and burnt to ashes, how can it ever return, therefore enjoyment 'here and now' is the goal of living in this school of thought¹¹.

The salient features of the Cārvāka school of philosophy are summarised in Figure 3.8.

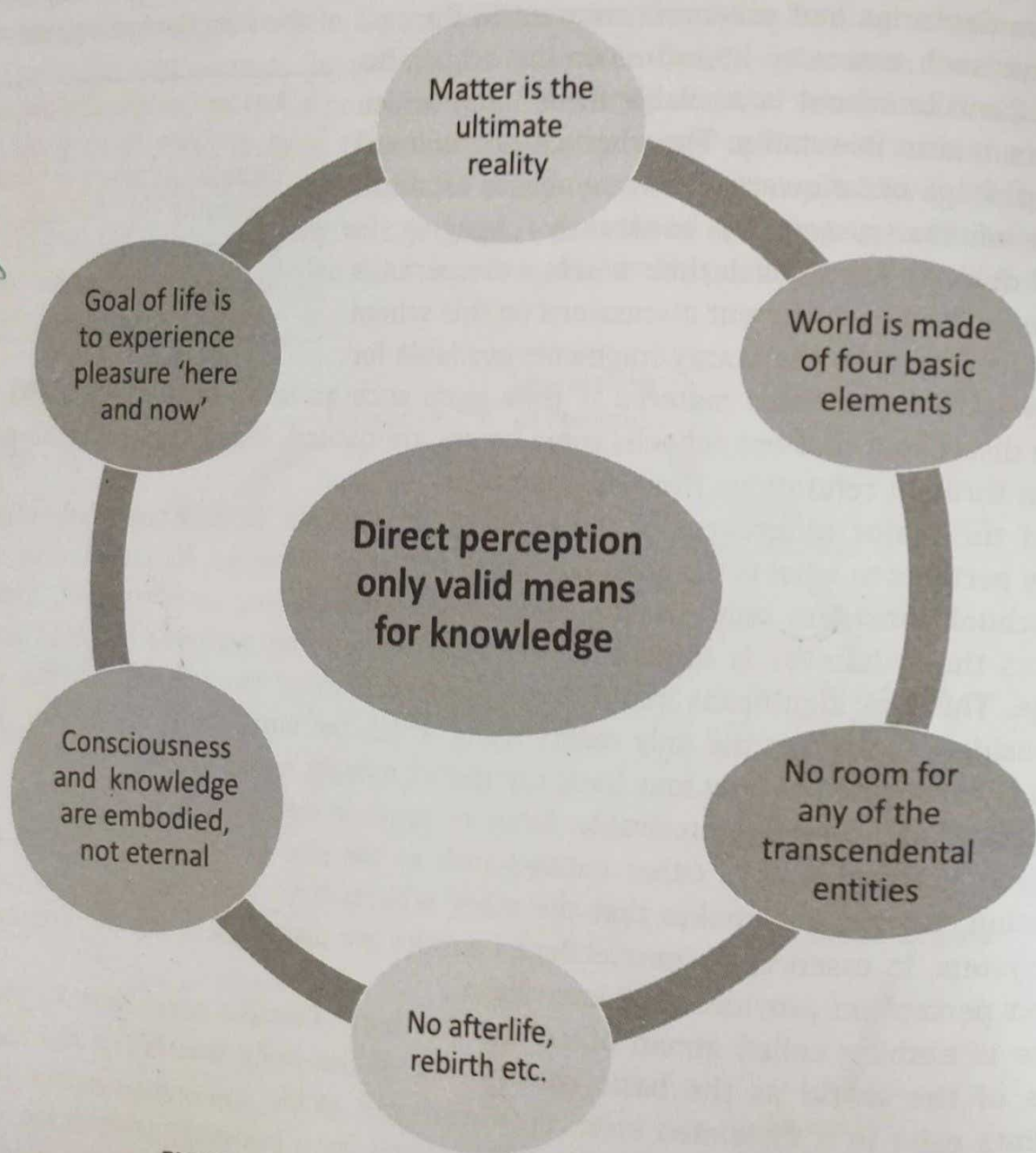


FIGURE 3.8 Salient Features of the Cārvāka School

- ▶ The word darśana is derived from the Sanskrit root, drś, 'to see', suggesting that these philosophical systems provide a true worldview and a vision for life and help us resolve the issues that we face in our life.
- ▶ The philosophical inquiry in India gave rise to several schools, each developing its own understanding of the world. Each of these schools presents its view with rigorous intellectual exercise and uncompromised importance on the self-experience.
- ▶ While the Hindu schools of philosophy rely on the Vedic text as the ultimate authority and means for valid knowledge, the Jaina philosophy relies on their canonical texts.
- ▶ The schools differ in presenting how to get the right knowledge and use it as the means for the ultimate liberation of the soul. The context for the philosophical discussion is the three inter-related concepts of God–Universe–Individual.
- ▶ According to the Sāṃkhya system, two basic elements constitute everything in this world, matter (Prakṛti) and spirit (Puruṣa). Puruṣa is the pure consciousness, sentient, changeless, eternal, and passive. Prakṛti on the other hand is the root cause of all activities including the entire creation.
- ▶ The unique thing that establishes Yoga as distinct among the other darśanas is that its emphasis on understanding the mind, its various states, its cognitive activities, and methods to control it. Yoga provides a structured and practical eight-step process to gradually reach a stage of complete cessation of the activities of Citta.
- ▶ Both Nyāya and Vaiśeṣika set out in their journey of the 'knowable' and define various categories to describe the same. The first is about the 'ways of knowing the reality' and the second is about the 'objects in the reality that is knowable'.
- ▶ The Nyāya system, placed enormous emphasis on the means of obtaining 'right knowledge'. Therefore, the unique contribution of Nyāya

school is its detailed inquiry of knowledge (Pramā) and valid cognition and its means (Pramāṇa).

- ▶ The main aim of Pūrva-mīmāṃsā school is to ascertain the meanings of the Saṃhitā and Brāhmaṇa portions of Veda which lay importance on the karma, performing rituals, and thereby attaining dharma as a means for liberation.
- ▶ The Vedānta schools of philosophy rely on three major texts: Brahma-Sūtras, Bhagavadgītā, and the Upaniṣads, known as prasthānatraya, for establishing their tenets.
- ▶ Śaṅkara propounded a monistic philosophy, known as Advaita, with the conception of the attribute-less God (Nirguṇa-brahman) as the ultimate reality.
- ▶ Matter, Jīva and Īśvara are three entities recognised in the Viśiṣṭādvaita school. Though all the three are ultimate, the first two have an absolute dependent relationship with Īśvara.
- ▶ The Dvaita school clearly admits two independent and mutually irreducible substances that make up the Universe: the Jīva and the Īśvara.
- ▶ Several concepts of the Jain school of thought is in line with the Vedic schools of philosophy. With syād-vāda, Jain philosophers are able to analyze claims made by various systems of thought and show them to be relative assertions of the truth as understood by the Jain tradition.
- ① ▶ Buddha's philosophy focuses on the means of ending the suffering of the individuals. It is based on four noble truths. The ultimate goal in Buddhist philosophy is to reach Nirvāṇa. Nirvāṇa is not a state reached after death, but something that is attainable in this very life.
- ② ▶ Cārvāka school of thought closely maps to the trait of materialism, which emphasises a life of enjoyment based on certain principles and assumptions about life. The concept of rebirth and other world is completely dismissed in this system.

Pañcatantra – A Treatise on Statecraft through Stories

Stories provide a powerful medium through which one can impart education on a wide range of topics. In modern parlance, stories are used as the basis for discussing several real-world problems. In business schools, case study methodology employs stories to analyze complex problems and various decision options. A shining example of the use of stories for education is Pañcatantra, a collection of stories involving the animal kingdom as the main characters. The text was composed by Viṣṇu Śarman.

The genesis of this text provides interesting insights into the power of storytelling in bringing the accumulated wisdom to use. Once a king named Amaraśakti had three sons who were ill-minded and were disinterested in learning. The king engaged a teacher, Viṣṇu Śarman to make his sons educated and eligible in governance at the earliest.

As the sons of kings were not interested in studying the śāstras Viṣṇu Śarman discovered a unique way to teach through stories. It had five Tantras (books), which is a set of stories on Governance, life, ethics, and morals, popularly known as Pañcatantra. It is also considered an important text on Nīti-śāstra.

Pañcatantra is considered as a Nīti-śāstra' as it deals with wise conduct in life. It consists of several stories nested within stories. Most of the characters are animals, who talk to each other and conduct several chores of life. Human characteristics, behaviour patterns, and even ethical values are ascribed to these animals. The wisdom in the five books deals with the following:

- ♦ *Book 1: The Loss of Friends (Mitra-bheda)* consists of a series of fables describing the conspiracies and causes that lead to close friends breaking up.
- ♦ *Book 2: Winning of Friends (Mitra-lābha)* emphasises the importance of friendships, teamwork, and alliances.

- ♦ *Book 3: Crow and the Owl (Kākolūkīyam)* contains eighteen fables discussing certain aspects of war and peace. It suggests that a battle of wits is a more potent force than a battle of swords.
- ♦ *Book 4: Loss of Gains (Labdha-praṇāśam)* presents some examples and consequences thereby highlighting actions to avoid.
- ♦ *Book 5: An ill-considered action (Aparīkṣita-kāraṇam)* presents twelve fables about hasty actions and jumping to conclusions without establishing facts and proper due diligence. Unlike the previous four books, the characters in book five are human beings.

The first book is the longest covering nearly half of the content and the last two books together constitute about 7% of the total.

The charm of Pañcatantra lies in its ability to frame complex matters of statecraft into stories involving nīti and daṇḍa from the Arthaśāstra and presenting them in easily relatable fables. For example, in Book 3 concerning the war of crows and owls, six acts of royal policy, i.e. Śāḍguṇya (peace, war, change of base, entrenchment, alliance, and duplicity) discussed in Arthaśāstra are quoted.

Much of the wisdom of ancient Indian statecraft which we are aware of today is attributable to Arthaśāstra and Pañcatantra, seems to have derived its inspirations and main messages from it. It is an intelligent rendering of Arthaśāstra using a pedagogical device consisting of a didactic story-based tool.

The captivating style of storytelling adopted by Pañcatantra has attracted people far and near. It appears that there is a version of Pañcatantra in every major language of India, and there are several versions of the text in more than 50 languages around the world.

Human beings face several challenges and moral and ethical dilemmas in life and constantly look for guidance to resolve them. These issues are eternal, not problems that only the current day society faces. Therefore, in our search of the answers to these, it is desirable to look back and understand how the ancestors addressed them. Every society develops a body of knowledge over time shaped by the accumulated experiences of the people, the inferences made out of

situations faced and experiments conducted. These experiences and insights are shared and stored for posterity through literary works. These showcase the cultural practices, history, science and technology, social customs, assumptions about life, and know-how prevailing at the time the literary work was created. It also describes how some of the challenges faced in personal and public life were addressed. It also provides useful guidelines to society on several issues faced in life. Therefore, the accumulated body of literature serves as the repository of wisdom through the ages. With a long civilizational history, India has a rich body of knowledge and experience that has developed within the society from time to time, captured through a variety of literary works.

In this chapter, we shall have a glimpse of the ancestral wisdom recorded in various literary works. Purāṇas and Itihāsas mainly belong to this category. The origin of the purāṇas and the itihāsas is not yet accurately established, but it dates back to antiquity. However, in the recorded history, several literary works have been authored in Sanskrit and other regional languages.

4.1 PURĀṆAS – AN ENCYCLOPAEDIC WORK

In the history of Indian literature, Purāṇas occupy a unique position. Purāṇas are literary sources of ancient and medieval Indian history and culture. The larger purāṇic repository provides valuable insights into the history of philosophy and religion and is a storehouse that provides deep insights into all aspects and phases of the society, culture during the ancient and medieval times in India. Purāṇas may be described as a popular encyclopaedia of ancient and medieval religious, philosophical, historical, personal, social, and political aspects of the society.

On account of the diversity of people and issues discussed in the purāṇic repository, purāṇas have significantly contributed to maintaining cultural pluralism in India. They played a role in influencing classical art forms such as dance and music and promoted social practices such as the celebration of various festivals. The spread of the cultural heritage among the masses has to a large extent possible only because of the purāṇas. Moreover, they have played a pioneering role in motivating and practice of ethical and moral values in the life of the people of Indian society using stories as a powerful medium to communicate the ideals. In this manner, Purāṇas have provided means by which socio-cultural values can blend with religious values and dharmic living.

- ♦ Purāṇas are one of the most important literary sources of ancient and medieval Indian history and culture.
- ♦ Purāṇas are companion texts to help understand and interpret the Vedas.

The literary beauty and story-based narration of ideas through a living tradition called 'Hari Katha' have perennially attracted the masses towards the purāṇas. On account of this, purāṇas enjoy a unique position in the sacred literature of ancient India. They closely align with the epics in form and substance. Purāṇas enable us to know the true import of the ethos, philosophy, and religion of the Vedas and are companion texts to help understand and interpret the Vedas. Therefore, knowledge of purāṇas is considered very important as evident from the following observations in some of the purāṇas¹:

- ✓ ♦ Skanda-purāṇa – Śruti (Vedas), Śmṛti (Rāmāyaṇa, Mahābhārata, Manu-smṛti, etc.) and Purāṇas are the three eyes of a scholar. One who sees the world with his three eyes is indeed a part of the supreme God.
- ✓ ♦ Viṣṇu-purāṇa – One who is a scholar of all the four Vedas, with its limbs and Upaniṣads in it and has no knowledge of purāṇas cannot be ever considered as a complete scholar.

As in the case of the knowledge repository in ancient India, purāṇas have been handed down the ages through oral transmission. Therefore, it is difficult to establish the period to the authorship of the purāṇas as there is no explicit mention. The current version available to us may be the work of several authors over the centuries.

However, using the information available in the purāṇas, Indology researchers have made some guesstimates. Accordingly, some of the extant Mahā-purāṇas, e.g. Vāyu, Brahmāṇḍa, Viṣṇu, Matsya, and Mārkaṇḍeya are assignable to a period of 300 CE – 660 CE, while some others

- ♦ The study of purāṇas and the upa-purāṇas are critical to developing an understanding of the social and religious institutions in India.
- ♦ Purāṇas deal with the evolution of the Universe, the recreation of the Universe, genealogies of Gods and seers, and the history of royal families.

such as Agni, and Garuda are assignable to a period of 600 CE – 900 CE and some others are assigned to a period later than 900 CE². On the other hand, in the Indian tradition, the antiquity of purāṇas is established using the available material in the purāṇas itself. According to Matsya Purāṇa, Brahmā remembered the purāṇas even before the Vedas³. We find mention of the word Purāṇa in the Vedas and Upaniṣads. For instance, in Chāndogyopaniṣad it is mentioned that Purāṇas can be considered as the fifth Veda⁴.

4.1.1 Mahā-purāṇas, Upa-purāṇas and Sthala-purāṇas

The purāṇic repository can be broadly divided into three: Mahāpurāṇas (Major Purāṇas), Upa-purāṇas (Minor Purāṇas), and Sthala-purāṇas (Regional Purāṇas).

- ♦ **Mahā-purāṇas**, the largest ever composed literature in the history of mankind is very extensive. There are 18 mahā-purāṇas⁵ and they are estimated to contain among them about 400,000 verses. The text in the purāṇas is in the form of several stories, anecdotes, clarificatory concepts, and rules and observances in a narrative style, often in the form of a reply to some issues raised during a dialogue between the narrator (usually a sage) and the audience (usually a group of sages, king, etc.). It enumerates stories of different forms of God, various kings, ṛṣis, etc. Usually, each purāṇa focuses on one form of the trinity (Brahma, Viṣṇu, Śiva) and celebrates the chosen trinity by narrating events centred around the trinity. For instance, Bhāgavata-purāṇa and Viṣṇu-purāṇa hold Viṣṇu as a supreme being, and Liṅga-purāṇa holds Śiva, etc. Some major purāṇas have discussed in detail a focused theme. For example, the Skanda, Padma and Bhaviṣya purāṇas deal mainly with pilgrimage to holy places (Tīrta-māhātmyas). The purāṇas cover a wide range of issues including the creation process and the geographical details, various rituals and their importance, certain scientific aspects, etc.
- ♦ **Upa-purāṇas** are similar to the mahā-purāṇas but are smaller in size. The number of Upa-purāṇas cannot be established with certainty, though about 100 of them can be enumerated from different sources. However, it is generally believed that there are eighteen of them which are prominent⁶. The list of upa-purāṇas differ in the verses found in various purāṇas⁷. So, it is difficult to list the names of eighteen upa-purāṇas exactly.
- ♦ **Sthala-purāṇas** are a class of sacred works which provide a connection between the land and the Divine in many forms. Each sthala-purāṇa is focused on a specific geographical place (usually a famous temple in a city) and weaves information centred

around the presiding deity. Various local cultural practices, customs, festivals, and other social practices form part of the narration. There are many Sthala-purāṇas, in different languages of India, most of them in Sanskrit.

Figure 4.1 graphically illustrates the purāṇic repository and its key features.

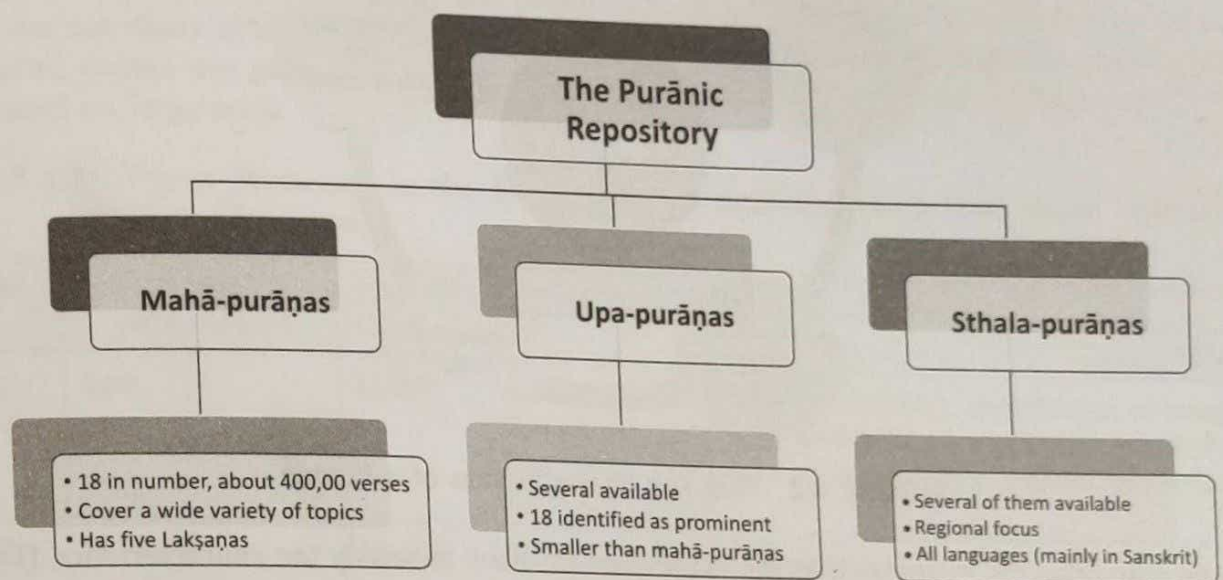


FIGURE 4.1 The Purāṇic Repository

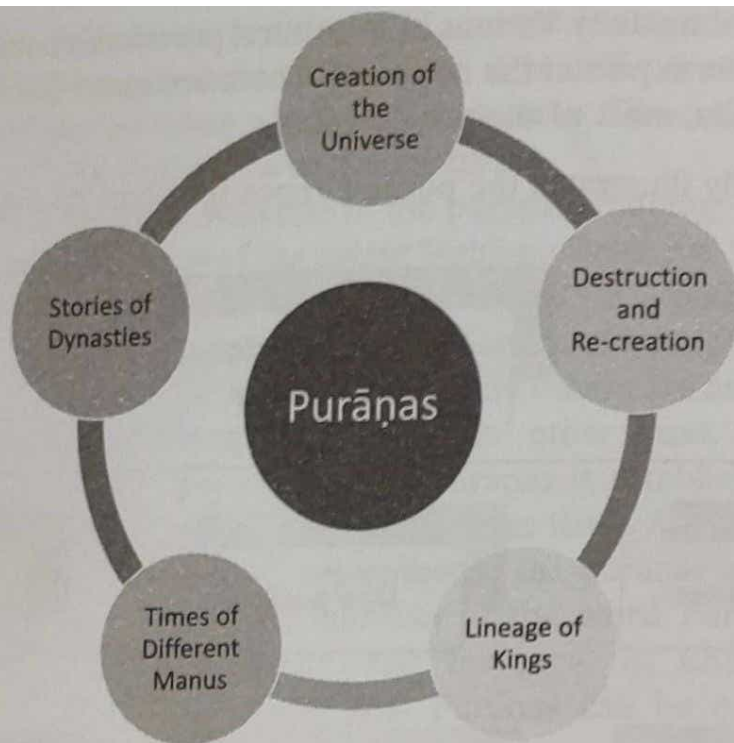


FIGURE 4.2 Five Key Characteristics of a Purāṇa

An expanded list suggests that the Mahā-purāṇas have to satisfy ten characteristics, (Daśa-lakṣaṇas), which additionally includes means of livelihood (vṛtti), protection of the universe (rakṣā), final emancipation (mukti), jīva, unmanifest (hetu), and Brahman (apāśraya). Besides the ten characteristics, the Purāṇas deal with the glorification of Brahmā, Viṣṇu, Sūrya and Rudra, and the four puruṣārthas (dharma – righteous conduct and living, artha – economics and polity, kāma – desire, and mokṣa – emancipation). The Purāṇas also contain, besides, a good deal on supplementary topics of religious instruction, rituals, dāna, vratas, bhakti, yoga, and also medicine, music, grammar, poetics, metrics, dramaturgy, astronomy, and astrology, architecture, sculpture, iconography, polity, raja-dharma, etc. Table 4.1 provides a compact summary of the issues discussed in the 18 mahā-purāṇas other than those covered under the purāṇa-lakṣaṇas described above.

4.2 ITIHĀSA AS A SOURCE OF WISDOM

The etymology of the word 'Itihāsa' provides us some clues about the nature of the work and its contents. It consists of three components, Iti + ha + āsa (verily did exist thus). For example, 'Rāma lived thus', 'Rāvaṇa lived thus', 'This happened thus', etc. The literature which explains such examples is considered as Itihāsa. This gives us a sense that the word Itihāsa points to 'history'. However, other definitions convey a wider meaning:

- ♦ The text which teaches the four goals of life Dharma, Artha, Kāma, and Mokṣa, and which deals with the stories of historical happenings is called Itihāsa¹⁸.
- ♦ The text explains the story of the past¹⁹.
- ♦ According to Kauṭilya – history (*Purāṇa*, *Itivṛtta*), tales (*Ākhyāyikā*), illustrative stories (*Udāharaṇa*), *Dharmaśāstra*, and *Arthaśāstra* are (known by the name) *Itihāsa*²⁰.

Therefore, Itihāsa is not merely a collection of stories related to some past events but an attempt to see the events through the lens of the four puruṣārthas. Though several literary works are considered as Itihāsa, only Rāmāyaṇa and Mahābhārata, also known as the great epics, have earned the fame of two major Itihāsas. At a cursory glance, the separation as Purāṇas and Itihāsas may appear artificial as both focus on narrating stories related to past events. However, there are crucial differences between the two:

- ♦ Itihāsas are narrations of the story in which the narrator has been part of the story. Thus, Mahābhārata authored by Vyāsa is a story in which he was also one of the actors, living at the time. So is the case of Vālmīki in Rāmāyaṇa.
- ♦ In this sense, Itihāsas are not stories about a distant past, as in the case of purāṇas. Therefore, there is a certain element of historicity to the Itihāsas, and these have been dated. On the other hand, establishing the period for the events narrated in purāṇas is not possible as they belong to a very distant past.
- ♦ In the case of purāṇas, there are five lakṣaṇas, as we have already seen. This involves discussion of multiple lineages of royal dynasties and ṛṣis. In contrast, in the two Itihāsas, there is a clear focus respectively on a single dynasty. In the case of Rāmāyaṇa, the focus is on the 'solar dynasty' (Sūryavaṃśa) and in the Mahābhārata, the focus is on the 'lunar dynasty' (Candravaṃśa). While the main flow of the story revolves around these respective dynasties, there are narrations of some purāṇic stories in Itihāsas, more as a side story to illustrate some point.

- ♦ By design, itihāṣas are not meant to be encyclopaedic in nature, as in the case of purāṇas. Instead, the focus is to narrate the events using the lens of the four puruṣārthas, as mentioned already.

4.2.1 Uniqueness of the Two Epics

The two Itihāṣas, Rāmāyaṇa and Mahābhārata occupy a very unique position in the repository of the Indian knowledge system. According to Swāmī

- ♦ Itihāṣa is stories related to some past events viewed through the lens of the four puruṣārthas.
- ♦ Rāmāyaṇa and Mahābhārata are the two encyclopaedias of the ancient Aryan life and wisdom, portraying an ideal civilisation.

Vivekānanda, "The Rāmāyaṇa and the Mahābhārata are the two encyclopaedias of the ancient Āryan life and wisdom, portraying an ideal civilisation, which humanity has yet to aspire after²¹". It is a common belief in Indian society that there is nothing that is not covered in these two Itihāṣas. Several poetries, texts on governance, and Nīti śāstra have been inspired by these two epics.

These two great epics have deeply influenced Indian society for time immemorial as is evident from several references to the heroes of the two epics in local matters throughout the length and breadth of the Indian sub-continent. In several sthala-purāṇas and local traditions we find examples such as, 'This rock contains the footprints of Rāma, the mountain got split into two unable to bear the weight of the Hanumān, the colour of this river is like this because mother Sītā had washed her clothes here, this is the cave where the Pāṇḍavas spent their Vanavāsa' etc. Another indication of the great influence of the two epics on society is evident from the fact that classical Indian literary works (mahā-kāvya) selected the themes from the two epics. For example, Kālidāsa's Raghuvamśa, Kumāradāsa's Jānakī-haraṇa, and Bhaṭṭi's Bhaṭṭi-kāvya (Rāvaṇa-vadha) were based on Rāmāyaṇa. Similarly, Kālidāsa's Abhijñāna-śākuntala, Bhāravi's Kirātārjunīya, Māgha's Śīsupāla-vadha and Śrīharṣa's Naiṣadhīya-carita were based on Mahābhārata. Similarly, Indian drama, dance, movies, and other performing arts have been greatly influenced by the two epics.

The two epics have taken different approaches to the issues of life and the propagation of dharma. Interestingly, both address the core issue of dharma and one's duty towards personal, family, and societal obligations. However, by virtue of the differences in their approach, they arrive at different answers to the questions related to the core issue. On account of this, they together serve as a complete handbook for sensible and successful living. Rāmāyaṇa's approach is simple and straightforward. Despite personal sufferings, the duty is clear, there is little scope for ethical and moral dilemmas. Presenting an ideal picture of life, characters, relationships, and values to be pursued, it seeks to elevate an individual to a higher plane. This approach may be called a 'normative approach' to life (essentially suggesting what ought to be done in life).

On the other hand, Mahābhārata has taken what could be referred to as a 'descriptive approach' (essentially illustrating what the reality of life is). In this approach, by presenting the actual happenings, and the moral and ethical dilemmas that different actors faced, the epic showcases the decisions made by the actors and their consequences in the long run. Rather than presenting the ideals of life, it shows life as it unfolds with all its attendant challenges. Instead of informing us what our duty is it merely presents using several anecdotes and nested stories of how various people discharged their duties and what the fallout was. The richness and the value of Mahābhārata lie in the fact that it is full of unsolved ethical riddles

and dilemmas, which are faced by ordinary human beings, wherein the characters struggle to find what is right, make mistakes and suffer the full consequences.

It is remarkable to notice that the two epics are not a mere piece of poetic or literary work but a living entity deeply engaging the lives of the Indian people, not merely of the intelligentsia, but also all sections of the society.

4.3 RĀMĀYAṆA – KEY ISSUES AND MESSAGES

Rāmāyaṇa consists of seven books which are called as Kāṇḍas, organized into 645 sargas (chapters), and 23,672 verses²². Rāmāyaṇa is a vast history of Rāma but also consists of many other related instances. In India, we have more than 40 versions of Rāmāyaṇa in different languages like the Rāmacaritmānas in Avadhi, Kamba-Rāmāyaṇa in Tamil, Mādhava-Kaṇḍali in Assamese, Goṇa Budha's Raṅganātha-Rāmāyaṇa in Telugu, Jagamohan-Rāmāyaṇa in Oriya, Narahari-Rāmāyaṇa in Kannada, etc. The story of Rāma is very well known in most parts of the world. It has been directly written in various languages in different countries. The Vālmiki-Rāmāyaṇa is translated into many languages. There are reportedly around three hundred versions of Rāmāyaṇa.

- ♦ Rāmāyaṇa takes a 'normative' approach while Mahābhārata takes a 'descriptive' approach to present events and decisions.
- ♦ Indian drama, dance, and other performing arts have been greatly influenced by the two epics.

The Rāmāyaṇa has played a significant role in cultural transmission between Indian and other Asian countries. It has travelled from India to other parts of Asia such as China, Turkestan, Burma, Thailand, Java, Cambodia, and Vietnam during the first millennium in the common era. It is interesting to note that almost all countries of Southeast Asia have the Rāma story, albeit in varied forms and content. From the thirteenth century onward, several Thai kings assumed the title Rāma, and the tradition continues. According to Rāmāyaṇa, Rāma is not merely a human being but an embodiment and a living entity of the concept of Dharma. Through several characters, Rāmāyaṇa provides a rich opportunity to understand important aspects of peaceful co-existence and successful living.

Bāla-kāṇḍa deals with the birth and childhood of Rāma and his brothers. Viśvāmitra takes along with him Rāma and Lakṣmaṇa through the forest and eventually to Mithilā leading to the marriage of Sītā to Rāma. One of the key messages that we can glean from this section is the importance of knowing the reality of life in the country by the king-to-be. Rāma's journey with Viśvāmitra through the forest and villages provided an opportunity for him to develop this awareness. This reminds us of an important principle stated in modern management parlance. In organisations, great leadership requires a good knowledge of the issues and realities facing the organisation. This is obtained only when the leader can connect to the people and situations in the organisation physically and mentally.

Ayodhyā-kāṇḍa describes the fast turn of events in which an attempt to coronate Rāma as the next heir apparent to the throne by Daśaratha was aborted and instead Rāma was sent to the forest for 14 years. The desire of Kaikeyī, the mother of Bharata, to install him as the prince went futile and Daśaratha died of shock and grief. This part of the epic provides an important message to the young that life can be a roller coaster ride. There is a need to understand that events in everyone's life can turn adversarial at any time. Rāma demonstrates that to be successful in life one needs to develop mental equanimity to handle such unforeseen events

in one's life. The conversation between Bharata and Rāma in the final portions of this section contains a wealth of information on public policy and administration and larger issues of life. We shall see some aspects of this in Chapter 14 of the book.

Aranya-kāṇḍa marks the large-scale elimination of demons in the forest in response to the request from the sages and seers, who were constantly harassed by these people. It also portrays the forceful and deceitful kidnapping of Sītā by Rāvaṇa. The epic reminds us that goodness needs to be protected from evil forces, lest it becomes unsustainable. The conversation between Mārīca and Rāvaṇa brings out an important message that while removing a poisonous snake that entered into a freshwater lake, the only source of livelihood for a village, 1000's of good fishes will die for no fault of theirs. It reminds us of the fact that an evil action by an individual will eventually trigger a bigger reaction from the system and in the process of eliminating the evil, several good aspects may also get destroyed.

Kiṣkindhā-kāṇḍa describes the efforts on the part of Rāma and Lakṣmaṇa to find Sītā, who was abducted by Rāvaṇa. The grief that they underwent was unparalleled as they were not even aware of what happened to Sītā and who took her away. The meeting of Sugrīva through the good offices of Hanumān instilled a ray of hope in their search. The epic brings out the importance of friendship, especially during periods of great distress. It also reminds us of the importance of wise counselling to wade through the challenges that we face in our life.

Sundara-kāṇḍa describes the heroic efforts of Hanumān to cross the ocean to locate Sītā and repose confidence in her. It also describes the horrifying state of affairs for Sītā and the grit and determination with which she faced the situation. The epic beautifully demonstrates the strength of the character of Sītā, with which she discovered unlimited courage within and faced the adversity with grace and purpose, which can inspire the womenfolk. Other messages include statecraft, dealing with ambassadors, and the presence of mind and tact required for assessing the strength of an opponent.

- ♦ Rāmāyaṇa encourages one to contemplate the difference between 'goodness' and 'greatness' and the need to have both to succeed in life.
- ♦ Mahābhārata discusses a variety of topics including law, philosophy, religion, and custom.

Yuddha-kāṇḍa describes the royal battle that erupted between Sugrīva's army and Rāvaṇa's army, eventually culminating in large-scale elimination of rākṣasas and their king Rāvaṇa as prophesied by Mārīca. Before the war, Rāvaṇa bluntly dismissed several attempts by his near and dear to instil a sense in him by pointing to the dangers of the imminent war and the need to broker peace with Rāma by giving up Sītā. Kumbhakarna while advising Rāvaṇa says, "One who takes the timely advice concerning Dharma, Artha and Kāma, about the pros and cons of the deeds from his scholarly councilors never gets into trouble²³". Rāvaṇa was very powerful but he flouted the advice given by his great councilors and well-wishers like Vibhīṣaṇa, Mārīca, Mālyavān, Maṇḍodari, and Kumbhakarna. This led to his downfall and death. The epic forcefully brings out the benefits of wise counseling and the dangers of hasty decisions arising out of an inebriated mind polluted with attachment to unreasonable and unethical desires.

Uttara-kāṇḍa has a collection of related stories not necessarily in chronological order as in the case of the previous books. It discusses certain events in Rāvaṇa's life not mentioned in the other books, the birth of Lava and Kuśa and the departure of Rāma and Sītā from this world.

The epic throughout depicts Rāma and Rāvaṇa in equal measure when it comes to 'greatness'. While describing the final encounter between Rāma and Rāvaṇa, Vālmīki remarks that it is impossible to find a simile to describe this, "Just as the vastness of the sky has no

simile and the majesty of the ocean also does not have a simile, in the same manner, there is no analogy to the war between Rāma and Rāvaṇa²⁴. Unfortunately, in terms of 'goodness' Rāvaṇa was no match to Rāma. The epic encourages one to contemplate the difference between 'goodness' and 'greatness' and the need to have both to succeed in life. Greatness without 'goodness' is a definite recipe for eventual disaster as demonstrated by Rāvaṇa. On the other hand, 'goodness' without 'greatness' may not enable one to cover much ground in his life.

Rāma lived as a human being and faced the day-to-day situations and conflicts of life from an ordinary human plane. This makes Rāmāyaṇa special and realistic. The Rāmāyaṇa describes situations where there are moral dilemmas and conflicts between two rights. Rāma was a perfect role model, who lived through the good and enjoyable aspects of life yet in a detached fashion. He was detached from power, wealth, greed, fame, and desire. He was kind and even minded to even those who opposed him. His ability to face difficult situations in life and to gracefully handle success and tragedy is an excellent example to follow.

The Vālmīki Rāmāyaṇa beautifully depicts the dharma of personal life, family life, and social order. The human characters in Rāma, Lakṣmaṇa, Bharata, Kausalyā, and Sītā bring out the principles of ideal living. It emphasises the ideal relationships between father and sons, between brothers, and between husband and wife. The concept of dharma as the way to a meaningful life could be a powerful theme for positive mental health. Many conflicts in life can be resolved using this perspective. These anecdotes, the conflicts, and the situations are the real nuggets of wisdom for the people.

4.4 MAHĀBHĀRATA – A SOURCEBOOK FOR WORLDLY WISDOM

Mahābhārata is one of the greatest poems in the world composed by Kṛṣṇa-Dvaipāyana (Veda Vyāsa). According to the available information, the original and first version of Mahābhārata which was named Jaya consisted of only about 8,000 verses and then the second version had 24,000 verses which was named Bhārata²⁵. It was later extended to up to 1,00,000 verses and was named Mahābhārata. There are multiple versions of the epic available. Therefore, a long-term project under the auspices of BORI, Pune started in 1919 under V.S. Sukthankar to prepare a critical edition of Mahābhārata. After collating information from 1,259 manuscripts and several years of work, a critical edition was finally brought out in 1966. The critical edition consists of more than 89,000 verses organised under 18 Parvas.

Multiple editions and redaction of the original text of Mahābhārata may appear to be a cause of concern. However, the leading incidents and characters of the epic have not been affected and modified by these changes. Mahābhārata depicts the trials and tribulations that

- ♦ Mahābhārata presents the reality of life and allows the reader to pick up their lessons for living.
- ♦ Nītis are the guidance to take a person to the path of Dharma by providing the right perspectives about life and life goals.

various kings of the lunar dynasty went through. To a large extent, the story focuses on the dispute between two families, the Kauravas and the Pāṇḍavas. The story unfolds steadily in the text but is interspersed throughout with a variety of topics including law, philosophy, religion, custom, and to some extent, geography and cosmography combined with several episodes and legends adding to the richness of the message. Table 4.2 concisely presents the broad

organisation of the text and the major topics covered in the text. The issues discussed in the text can be broadly summarized under five major heads:

✓ **TABLE 4.2** Overall Structure and Organisation of Topics in Mahābhārata

Parva	Chapters	Topics
Ādi	225	Stories of the ancestors of Kuaravas and Pāṇḍavas; birth and education of Kauravas and Pāṇḍavas; enmity in their childhood; Draupadī's marriage to Pāṇḍavas; Arjuna's marriage to Subhadra
Sabhā	72	Performance of Rājasūya yajña by Yudhiṣṭhira, the game of dice maneuvered by Duryodhana, and its tragic consequences
Āraṇyaka	299	Pāṇḍavas in exile, several stories from the past, famous conversation known as Yakṣapraśna
Virāṭa	67	The stay of the Pāṇḍavas incognito in the kingdom of Virāṭa, Virāṭa princess Uttarā gets married to Abhimanyu
Udyoga	197	Unsuccessful peace parleys and preparation for the war, the famous discourse of sage Sanatsujāta to Dhṛtarāṣṭra
Bhīṣma	117	Śrīmad <i>Bhagavad Gītā</i> , detailed descriptions of the first ten days of the war, Bhīṣma mortally wounded by Arjuna
Droṇa	173	The heroic exploits of Droṇa and his death through a planned strategy; Abhimanyu's tragic death
Karṇa	69	Death of Duśśāsana, the second of the Kaurava brothers, fall of Karṇa at the hands of Arjuna
Śalya	64	Final encounter between Bhīma and Duryodhana on the last day of the war
Souptika	18	The gruesome massacre of the Pāṇḍava army in the night during sleep by Aśvatthāman
Strī	27	Lamentations of the widows of the dead warriors
Śānti	353	Wonderful discourses on all aspects of <i>dharma</i> by the patriarch Bhīṣma at the request of Yudhiṣṭhira, Bhīṣma's demise, Yudhiṣṭhira's coronation
Anuśāsana	154	
Āśvamedhika	96	Departure of Śrī Kṛṣṇa to Dvārakā, Āśvamedha Yajña by Yudhiṣṭhira
Āśramavāsika	47	Departure of Dhṛtarāṣṭra to the forest along with Gāndhārī, and Kuntī; their subsequent death in a forest fire
Mausala	9	Mutual destruction of the Yādava heroes and also the death of Śrī Kṛṣṇa
Mahāprasthānika	3	Final journey of the Pāṇḍavas, their death on the way, Yudhiṣṭhira alone reaching heaven
Svargārohaṇa	5	
Harivamśa	118	Supplement of Mahābhārata
Total	2,113	

SUMMARY

- ▶ With a long civilisational history, India has a rich body of knowledge and experience that has developed within the society from time to time, captured through a variety of literary works.
- ▶ The purāṇic repository provides valuable insights into the history of philosophy and religion and is a storehouse that provides deep insights into all aspects of the society, culture, and living during the ancient and medieval times in India.
- ▶ The purāṇic repository can be broadly divided into three; Mahā-purāṇas (Major Purāṇas), Upa-purāṇas (Minor Purāṇas), and Sthala-purāṇas.
- ▶ There are 18 mahā-purāṇas and they are estimated to contain among them about 4,00,000 verses. The number of Upa-purāṇas cannot be established with certainty, though about 100 of them can be enumerated from different sources. There are many Sthala-purāṇas, in different languages of India, most of them in Sanskrit.
- ▶ The purāṇas contain several discussions related to the plant kingdom and its role in addressing food, medicine, and other requirements of society. Since one of the themes of the purāṇas is to describe the creation process, there are discussions on the stages of childbirth from the embryo.
- ▶ Several purāṇas have discussions on astronomy, which consists of ideas in the pre-telescope age. The purāṇas have material on geography as they discuss topics related to modern-day concepts such as cosmogony, cosmology, and cosmography.
- ▶ Itihāsa is not merely a collection of stories related to some past events but an attempt to see the events through the lens of the four puruṣārthas.
- ▶ The two epics, Rāmāyaṇa and Mahābhārata address the core issue of dharma and one's duty towards personal, family, and societal obligations, but have taken different approaches.
- ▶ The Vālmīki Rāmāyaṇa beautifully depicts the dharma of personal life, family life, and social order. The anecdotes, the conflicts, and the situations discussed in the epic are the real nuggets of wisdom for the people.
- ▶ Mahābhārata depicts the trials and tribulations that various kings of the lunar dynasty went through. The story is interspersed throughout with a variety of topics including law, philosophy, religion, custom, and to some extent, geography and cosmography.
- ▶ Nītis are the guidance to take a person to the path of Dharma by providing the right perspectives about life and life goals. They do not take the approach of storytelling but utilizes verses set in poetic metres to concisely communicate the message.
- ▶ Sāmānya-nīti focuses on the good aspects of living and provides helpful insights on the code of conduct, differences between a wise person and others, good and evil, and success and failure in life pursuits.
- ▶ Rāja-nīti focuses on the political and administrative tasks and provides ideas on several do's and don'ts for a king.
- ▶ The concise poetry which communicates the valuable message for the society with some interesting examples is called Subhāṣita. These are nothing but a systematic collection of insightful observations, examples, and principles of life from a wide range of existing repositories of literary works.
- ▶ The messages contained in the subhāṣitas are relevant to all the ages in the Indian tradition; these are specially taught to the children so that they can imbibe these ideas while they are young and can make use of this wisdom all through their life.

✓ 5.1 COMPONENTS OF A LANGUAGE

How many of us can read and comprehend Shakespeare's works or native Indian literature such as Rāmāyaṇa? Even if we know the language in which these texts are written, it is difficult because the way the language is used undergoes changes. Language is a tool used by everyone in a community and it is very difficult to maintain it unchanged. We also notice that there are differences in the same language spoken by people from different regions. For these reasons, the literature in a language becomes incomprehensible for the people in the future though they may be speaking the same language. Studying the structure of the language helps us not lose the underlying principles that govern a language and ensures that the received wisdom from the ancestors is not lost. It also maintains continuity in language processing.

- ♦ One of the Vedāṅgas known as Vyākaraṇa focuses on linguistics and phonetics aspects of Sanskrit language.
- ♦ Aṣṭādhyāyī is considered a fine creation of human intelligence and the best available descriptive model of a language.

Communication is key to trade, science and technology, and societal progress. It hinges on our ability to effectively process language as it is central to all human transactions and pursuits.

Language processing has two dimensions: receptive and productive. The *receptive* part of a language deals with the ability of an individual to receive language inputs from multiple sources and process them to decipher the intended message and comprehend them. On the other hand, the *productive* part of the language is to transmit back to others for their consumption. The focus in the former is on listening and reading, whereas it is on speaking and writing in the latter. Viewed from another perspective, sound (listening and speaking) and script (reading and writing) are the essential elements of a language. Therefore, language processing can be represented in a 2×2 framework as shown in Figure 5.1. Linguistics addresses all these aspects of a language. Phonetics will cater to the receptive and productive aspects of the sound and a syntactical structure will cater to the scriptural aspects of a language.

Linguistics is a branch of language research that provides a scientific study of a language. It is a systematic study of language to understand speech sounds, grammatical structures, and meaning. It helps us analyse the language form and meaning and identify systematic methods integral to the language to derive the word forms and their meaning using structured rules and syntax. The earliest approach to a systematic treatment of linguistics is attributed to the Indian

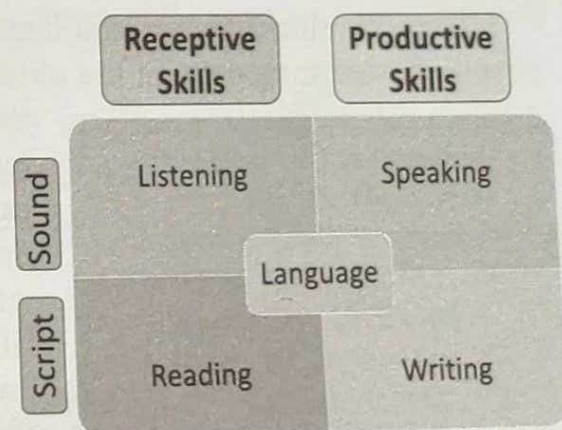


FIGURE 5.1 Components of a Language

grammarians, Pāṇini who lived in the 6th Century BCE. His work on Sanskrit grammar known as *Aṣṭādhyāyī* is a magnum opus on linguistics. In this chapter, we shall see more details of this.

5.2 PĀṆINI'S WORK ON SANSKRIT GRAMMAR

In the Indian context, since the preservation of the Vedas and their meanings was given utmost importance, linguistics was an important discipline to be studied by everyone. Therefore, three out of the six Vedāṅgas, Śikṣā, Nirukta and Vyākaraṇa focused on phonetical and syntactical aspects of language. We have briefly discussed these in Chapter 2 of the book. On account of this, people who are trained in Sanskrit can comprehend ancient works such as the epics, Purāṇas, and the Vedic corpus in a lossless fashion. Several Indian languages have borrowed the syntactical and phonetic structure, language organization, and vocabulary from Pāṇini's work on Sanskrit grammar. Therefore, it is important to know this in some detail. Pāṇini's work on Sanskrit grammar, known as *Aṣṭādhyāyī* is a process of refinement and syntactical structuring, because of which the language is called Saṃskṛta (the refined one) and is considered a fine creation of human intelligence.

The *Aṣṭādhyāyī* is the culmination of a long grammatical tradition. Pāṇini brought his originality into action and proposed a structure to the grammar. It is the best available descriptive model of a language. The greatness of this work is evident from the fact that it has overshadowed all previous attempts on Sanskrit grammar. Moreover, several commentaries written on it is another proof of its prominence in linguistics. In the 4th century BCE, Kātyāyana composed a commentary (vārtika) on the Pāṇinian work, which served to provide further explanation of the work and clarified certain aspects of *Aṣṭādhyāyī*. Another great work on *Aṣṭādhyāyī* is the *Mahābhāṣya*, a commentary on the *Aṣṭādhyāyī*, authored by Patañjali in the 2nd century BCE. These three are considered the three great sages (tri-muni) of Sanskrit grammar.

While framing a syntax and linguistic framework, the challenge is to find an efficient set of rules to accommodate all the observed patterns and variations in the use of a language by the society. If there are robust rules that govern a language

- ♦ Pāṇini composed 3,983 rules to accommodate all the patterns and variations in Sanskrit language.
- ♦ The basic approach of Pāṇini and its distinguishing features make Sanskrit a powerful language and eternal in its appeal.

leaving several exceptions, it is not useful. At the other extreme, to accommodate all the possible variations, the number of rules framed may become very large, again making it less useful. The greatness of the Sanskrit grammar laid out by Pāṇini lies in its astuteness in establishing a framework for linguistics that nicely balanced both these aspects. Pāṇini did not write a set of rules and insisted that

everyone must follow these rules to use the language. This is possible when we develop a computer language such as C or Python, for example. Pāṇini composed 3,983 rules (known as Sūtras) to mostly accommodate the patterns and variations in the Sanskrit language and arranged them in eight chapters (therefore, the name *Aṣṭādhyāyī*). Each chapter is further divided into 4 quarters, thereby making it 32 quarters in all.

The basic approach of Pāṇini and its distinguishing features make Sanskrit a powerful language and eternal in its appeal. These are summarized below:

- ♦ The entire vocabulary of the Sanskrit language could be created using the 3,983 rules. Few exceptions need special handling. The rules are aphorisms (known as sūtras), which are easy to commit to memory. What it implies is that if someone is familiar with

these rules and how it needs to be applied, then it amounts to gaining unambiguous mastery of Sanskrit language. The educational systems in India until the introduction of the Macaulayan system of education in the 19th century CE ensured this mastery for the students.

- ◆ Language processing and word generation are strictly rule-based and derivative in nature. Proper application of the rules must result in a valid word (a form of the language). There is no need to make any additional assumptions to derive any word.
- ◆ The derivation of words using the rules could be done using a step-by-step process. What it implies is that technically speaking, starting from a base (a verbal root or a nominal root), the logic can search if any of the 3,983 rules could be applied to the current transformation of the root. If the rule can modify, the current structure it will perform the operation. The procedure stops when none of the rules could be further applied. The result is the final word. This rule-based recursive structure to language is highly amenable to computer-based processing. The entire process is logical, unambiguous, and rational.
- ◆ The entire scheme for word generation follows a highly modular approach. Two basic components form a word. Each word is formed out of a base (verbal or nominal root). To this, one or more suffixes are added to generate the word. On account of this method of word generation, the Sanskrit language inherently has a high degree of 'patterns' of words and word structures that makes it very efficient to develop vocabulary.
- ◆ Since the entire language is rule-based, the vocabulary is not fixed or static for the language. The rules can be used for generating new words, as long as the rules are not violated. Therefore, the language is dynamic, can construct new words as demand arises, and can maintain its relevance. This has implications for lexicographic studies in the language.
- ◆ Aṣṭādhyāyī deploys several interesting data structures and computational elements that make it unique among linguistic studies. 7

5.7 IMPORTANCE OF VERBS

Language is required only when we are in action. If no one is engaged in action, then there is no need for language. Therefore, several aspects of linguistics revolve around the action. The discussion in the previous section establishes the importance of a verb (kriyā-pada) in Sanskrit language. The importance of the verbs is further amplified by the fact that several of the noun roots are also derived from the verb roots only.

Since most words (both the verb forms and noun forms) originate from the verb roots (dhātus) there is another important dimension to this. In Sanskrit, we often find several synonyms for a word. Each synonym for a word is often derived from a dhātu. With a good knowledge of the dhātus, we will be able to make sense of the synonyms, and the unique meaning the word derives from a particular synonym. This significantly increases the power of the language as one can choose an appropriate synonym depending on the context in which the word is intended to be used. Let us consider an example to understand this idea. Let us consider the English word 'fire'. A common word in Sanskrit that may come to one's mind is 'Agni'. Amarakoṣa, a book of synonyms in Sanskrit provides 34 synonyms for the word Agni¹⁵. Let us take four synonyms from this list and understand the meaning of the synonym:

1. वह्निः (vahniḥ) – this is derived from the dhātu, 'वह् (vah)', meaning 'to carry'. Therefore, when we need to communicate 'fire' in a sense of a carrier (as in the case of āhuti being offered to the Gods in a sacrifice) this term may be appropriate.
2. पावकः (pāvakḥ) is derived from the dhātu 'पू (pū)' meaning 'to purify'. When we need to discuss the role of fire as a purifying agent, then the choice of this synonym becomes most appropriate.
3. शुष्मा (śuṣmā) is derived from the dhātu 'शुष् (śuṣ)' meaning 'to dry or shrink'. Fire plays the role of drying air, water, etc. or shrink organic objects by evaporating the water by virtue of the heat it possesses. In such contexts, the use of this synonym may be appropriate.
4. दहनः (dahanḥ) is derived from the dhatu 'दह् (dah)' meaning 'convert or burnt to ashes'. Where we need to communicate the power of fire to burn off objects, this synonym communicates the idea appropriately.

These examples demonstrate to us that the knowledge of the dhātus from which a noun has been derived is important to communicate an idea most appropriately.

Prefixes for Verb Forms

Most of the word generation as we have seen so far works on adding suffixes to the base and applying relevant grammatical rules to arrive at the final word. However, prefixes (upa-sargas), are appended to the verb forms in order to create additional words. There are 22 prefixes, and

and how it systematically developed through the ages. We shall dwell on these issues in this chapter.

6.1 NUMBER SYSTEM IN INDIA – HISTORICAL EVIDENCE

The contribution of ancient Indians to the development of mathematical concepts is well known and acknowledged. Many Scientists and writers from the West have time and again acknowledged this aspect, as evident from the following:

- ♦ Based on extensive research, Ifrah presented evidence from Europe and the Arabic world to show that modern number systems indeed originated in India. Using several references during 810 CE to 1814 CE, he presented several observations of the past writers on Indian Mathematics¹.
- ♦ Laplace remarked, "The ingenious method of expressing every possible number using a set of ten symbols (each symbol having a place value and an absolute value) emerged in India. Its simplicity lies in the way it facilitated calculation and placed arithmetic foremost amongst useful inventions"².
- ♦ In his book on India, composed around 1030 CE, Al-Biruni wrote, "Whilst we use letters for calculation according to their numerical value, the Indians do not use letters at all for arithmetic. And just as the shape of the letters that they use for writing is different in different regions of their country, so the numerical symbols vary"³.

A number of archaeological excavations supplement the linguistic evidence provided by the early writers about the maturity and supremacy of the Indian mathematical foundations. The street widths in the Indus-Saraswati Civilization were highly standardized. For instance, Kalibangam, a city in the Indus-Saraswati Civilization (in Rajasthan, India) had street widths of 1.8 m, 3.6 m, 5.4 m, and 7.2 m⁴. These were built to the standard dimensions of 1, 2, 3, and 4 Dhanus (this unit is discussed in the section on measurements) respectively. Such widths are found at other sites also. The excavations at Harappa, Mohenjo Daro, Dholavira, and Lothal show that several constructions were done using fired bricks of standard dimensions and fine geometrical and material quality. The dimensions of the bricks appeared to be standard with length $\times$ width $\times$ depth in the ratio 4:2:1. Weights and linear measures follow a similar pattern wherever they have been found. This similarity across sites spanning several hundred kilometres points to the use of a standard unit of measurement at that time, signalling mature mathematical thinking.

- ♦ Several archaeological excavations provide evidence about maturity and supremacy of Indian mathematical foundations.
- ♦ An inscription on a temple wall in Gwalior dating back to the ninth century CE is considered the oldest recorded example of a zero.

In the Arthaśāstra, there is a mention of two types of Dhanus as units for measuring lengths and distances: Dhanus = 96 Angulas; Gārhapatya-dhanus = 108 Angulas used for measurement of roads and distances. The oldest dated Indian document containing a number written in the place-value format that we use today is a legal document dated 594 CE from the Bharukachcha (or Broach) region in Gujarat. In an inscription at Gwalior dated 'Samvat 933' in the Vikrama calendar (876 CE) the numbers 50 and 270 were recorded with a small circle appearing at the appropriate positional place for zero. An ancient Indian scroll known as the Bhakshali manuscript was discovered in a field in 1881 giving evidence of the earliest recorded use of

zero. Carbon dating of this has revealed that it was probably written in the third or fourth century CE. An inscription found on a temple wall in Gwalior dating back to the ninth century CE is another recorded example of a zero.

6.2 SALIENT FEATURES OF THE INDIAN NUMERAL SYSTEM

The Indian numeral system has a long history. The origin and the evolution of the numbers could be traced from the time of the Vedic period. Being an oral tradition, unique and unambiguous names were to be attributed to the numbers. Therefore, as early as the Ṛgveda time, the use of numbers and unique names prevailed. The Sanskrit language has unique names for numbers starting from one and going up to very large numbers. The first nine digits have unique names (ekam, dve, trīṇi, catvāri, pañca, ṣaṭ, sapta, aṣṭa, nava). There are unique names for numbers from 10 to 100 in steps of ten (*daśa, viṃśati, triṃśat, catvāriṃśat, pañcāśat, ṣaṣṭi, sapṭati, aśīti, navati, śata*). Beyond 100, there has been the use of names for numbers, which extend up to very large numbers. There are many references in the literature pointing to the use of such number names.

Spread of Indian Mathematical Concepts Westwards

The catalogue of the Sui dynasty (610 CE), pointed to several Chinese translations of Indian works on astronomy and mathematics, which are no longer extant. Records from the seventh century (of the Tang dynasty), suggest that Indian astronomers were employed at the Astronomical Board of Chang-Nan to teach the principles of Indian astronomy and calendar. An Indian named Gautama-Siddhārtha was reported to have constructed a calendar along the lines indicated in the Indian *Siddhāntas*.

Indian numerals reached the court of second Abbasid Caliph al-Mansur (753–774 CE) from Sindh. Moreover, the Indian decimal place-value system reached at least a century earlier as evident from a passage attributed to Nestorian Bishop Severus Sebokht (662 CE). In a book written during 820 CE by al-Khwarizmi, the operations of addition, subtraction, multiplication, division and the extraction of square roots according to the Indian system was explained. His work was translated into Latin by 12th Century CE and there were several commentaries in Latin on this book.

In the transmission of Indian numerals to Europe, via the Arab world, Spain played an important role as it was under Muslim rule for many years. Documents from Spain and coins from Sicily provide evidence of the spread. Fibonacci (1170–1250 CE) learnt the Indian numerals during his travels in North Africa, Egypt, Syria and Sicily and wrote a book bringing these aspects into focus. The spread westwards continued slowly, displacing Roman numerals. It was simply a matter of time before the new numerals were put to use by the bankers, traders and merchants for their daily calculations.

Source: Adopted from **Joseph (2009)**. "A Passage to Infinity: Medieval Indian Mathematics from Kerala and Its impact", Sage, Chapter 8.

One can appreciate the importance of the Indian numeral system in scientific development if the following contributions are understood:

- ♦ A legacy of using large numbers with unique number names for these large numbers
- ♦ Developing a robust place value system for the numerals
- ♦ The concept of zero and its use beyond being a placeholder
- ♦ A decimal system that opened vast possibilities for arithmetic operations

6.2.1 The Concept of Zero and Its Importance

Today, zero, both as a symbol (or numeral) and a concept meaning the absence of any quantity allows us to perform calculus, solve complicated equations, and to have invented computer operations using binary digits. Among the significant contributions of the ancient Indians is the concept of zero. The available evidence shows that the concept of zero was established during the period 500–300 BCE and fully developed in India by 600 CE. Ancient Indians were able to use a decimal system that allowed them to develop a method for handling large numbers. In this process, the use of number zero became inevitable as a placeholder. The number name to indicate zero is Śūnya. Piṅgala a second century BCE Indian philosopher authored Chandaḥśāstra, which dealt with the metres used in Sanskrit poetry in which the word śūnya was used, which obtained the mathematical connotation of 0. It later became its proper name as a number.

- ♦ The concept of zero was established during the period 500–300 BCE.
- ♦ Ancient Indians used a decimal system that allowed them to develop a method for handling large numbers.

Brahmagupta developed a symbol for zero in 628 CE. With this invention, zero could be used as an independent numeral for computational purposes. The real power of zero was evident when the use of zero was beyond a mere placeholder. Ancient Indians used zero in computations ahead of others and thereby elevated zero from a placeholder to a numeral. This is considered one of the greatest breakthroughs in the history of mathematics. Bhāskara II in his Bija-gaṇita introduced the properties of zero when mathematical operators such as addition and subtraction are operated on it. He stated that its nature does not change when numbers are added to subtracted from zero.]

TABLE 6.3 Word Numerals for Bhūta-saṃkhyā System

Number	Represented by (partial list only)*
0	śūnya, anata, pūrṇa, kha
1	ādi, candra, prithivī, eka
2	aśvin, pairs of limbs, ayana, dvandva, dvi
3	rāma, guṇa, loka, kāla, agni, trinetra
4	veda, śruti, yuga, aśrama, varṇa, samudra, kṛta
5	bhūta, śāstra, bāṇa, pāṇḍava, indriya
6	aṅga, ṛtu, darśana, śaṇmukha, ṣaṭ
7	ṛṣi, adri, svara, dhātu, chandas
8	vasu, bhujaṅga, siddhi, dik, kuñjara, nāga
9	gr̥ha, aṅka, nanda
10	dik, angulī, avatāra, rāvaṇaśīras
11	rudra
12	āditya, rāśi
13	viśva, kāma
14	manu
15	tithi, dina
* For all the items listed, any word from the synonyms may be used	

6.3.2 Kaṭapayādi System

Another system to convert the numerals to words is to associate a number to one or more alphabets. Once this association is established, the numbers can be replaced with a corresponding alphabet. Using the alphabet in the place of numbers, one can construct words, which by deciphering one alphabet at a time will reveal the number expressed in the word. The advantage of such a system lies in representing long (or large) numbers using a nice word, which can be easily remembered. This is particularly very valuable in an oral tradition, where large numbers may have to be remembered and communicated without loss of digits. This provided a very efficient method of presenting results of complex calculations using number symbols, and then codifying and committing to memory effortlessly.

Kaṭapayādi system employs certain rules to convert the numerals to alphabets:

- The vowels when standing alone indicate the number zero. In all other cases when they are conjoined with consonants, they have no role in this system other than facilitating the pronunciation of the consonants when used in a word.
- Each consonant is uniquely associated with a number from 0 to 9. For example, the consonants ka-ṭa-pa-ya was used to denote number 1. Therefore, the system was referred to as the Kaṭapayādi system. Similarly, kha-ṭha-pha-ra denotes number 2, and so on. Table 6.4 has the details of the assignment of consonants to the numbers.
- As is evident from the table, more than one consonant may be associated with each of the numerals.
- When more than one consonant is used in conjunction, only the terminal consonant preceding a vowel is to be considered for the identification of the corresponding numeral.

- (e) If a consonant is not conjoined with a vowel and stands alone, then it will not be considered for identification of the numeral.
- (f) Similar to the bhūta-saṃkhyā system, the numbers are to be read as units - tens - hundreds ... etc.

TABLE 6.4 Letter Numerals of the Kaṭapayādi System

1	2	3	4	5	6	7	8	9	0
क (ka)	ख (kha)	ग (ga)	घ (gha)	ङ (ṅa)	च (ca)	छ (cha)	ज (ja)	झ (jha)	ञ (ña)
ट (ṭa)	ठ (ṭha)	ड (ḍa)	ढ (ḍha)	ण (ṇa)	त (ta)	थ (tha)	द (da)	ध (dha)	न (na)
प (pa)	फ (pha)	ब (ba)	भ (bha)	म (ma)					
य (ya)	र (ra)	ल (la)	व (va)	श (śa)	ष (ṣa)	स (sa)	ह (ha)		

The use of conjoined vowels and consonants may be to aid generate pleasing and easy to remember words while representing a number, but the logic of identification of the number is based on the above rules. The origin of this system is traced back to the 5th century CE. Before the advent of the Kaṭapayādi system, the bhūta-saṃkhyā system was popular. Although this system is much simpler and easy to use, it was not in general use since there were four variations of the system. Table 6.4 has the letter numerals pertaining to the kaṭapayādi system.

Mādhavācārya's Approximation to π

Ancient Indian mathematicians have provided multiple approximations to π . Many of these mathematical details, numbers and computations required are invariably provided in the form of some verses using word numeral systems such as bhūta-saṃkhyā system. One such approximation is due to the 11th Century CE mathematician Mādhavācārya. The verse below contains the relevant details. Let us use the bhūta-saṃkhyā system to decipher the details.

विवुधनेत्रगजाहिहताशनत्रिगुणवेदभवारणबाहवः ।
नवनिखर्वमिते वृत्तिविस्तरे परिधिमानमिदं जगदुर्बुधाः ॥

vibudha-netra-gaja-ahi-hutāśana-triguṇa-veda-bha-vāraṇa-bāhavaḥ /
navanikharva-mite vṛttivistare paridhimānam-idam jagadurbudhāḥ //

In the first line of the verse, numbers are provided using the bhūta-saṃkhyā system. Let us first extract the numbers. *vibudha* - Devas (33); *netra* - Eyes (2); *gaja* - Elephant (8); *ahi* - Snake (8); *hutāśana* - Agni (3); *tri* - (3), *guṇa* - (3); *veda* - (4) *bha* - stars (27); *vāraṇa* - Elephant (8); *bāhu* - Hands (2); Therefore, the number mentioned is 2,827,433,388,233. In the second line of the verse, there is again a mention of a number; *nava* - (9); *nikharva* - 10^{11} . This number is 9×10^{11} . The balance part of the śloka mentions that this is the ratio of the circumference (*paridhimānam*) to the diameter (*vṛttivistare*) of the circle. Taking this ratio will yield us the value of π .

$$\text{The value of } \pi = \frac{2827433388233}{9 \times 10^{11}} = 3.14159265359222$$

Source: Based on <https://www.youtube.com/watch?v=Ke9yxVsUPvo&list=PLbMVogVj5nJThf31TNSQzuN7zqxe7HdRN&index=6>.

TABLE 6.9 Eight Ganas Defined by Pingala

Sl. No.	Gana Name	Binary Word*
1	'Ya' Gana	100
2	'Ma' Gana	000
3	'Ta' Gana	001
4	'Ra' Gana	010
5	'Ja' Gana	101
6	'Bha' Gana	011
7	'Na' Gana	111
8	'Sa' Gana	110

* 1 = Laghu; 0 = Guru

There is an easy to remember mnemonic for the composition of the eight *gaṇas* given by the following short phrase:

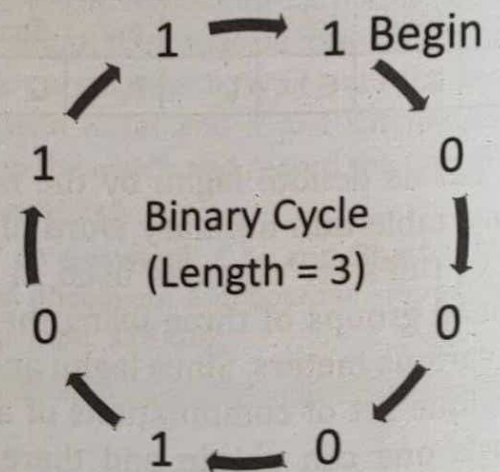
यमाता-राज-भान-सलगम्
yamātā-rāja-bhāna-salagam

Analysing this phrase using the rules of *laghu* and *guru* and converting it into a binary word, we get the result as presented in Table 6.10.

TABLE 6.10 Laghus and Gurus in the Phrase

ya	mā	tā	rā	ja	bhā	na	sa	la	gam
L	G	G	G	L	G	L	L	L	G
1	0	0	0	1	0	1	1	1	0

If we examine the binary word representation of the mnemonic, we notice that the last two digits, '10' is repeated. If we remove this repetition, the remaining eight digits of the binary word form a cycle for a 3-digit binary number, as indicated in Figure 6.2. Using this binary cycle, we can generate all the 8 unique combinations of the binary word of length three that constituted the eight *gaṇas*. For example, from the point indicated as 'begin' in Figure 6.2, we can progressively derive all the 8 *gaṇas* and their binary composition taking three consecutive binary numbers at a time. '1 - 0 - 0' forms the 'ya' *gaṇa*, '0 - 0 - 0' forms the 'ma' *gaṇa* and so on. Very recently, this method of string generation was discovered using combinatorial mathematics known as the De Bruijn sequence. It is a **sequence** of order '*n*' on an alphabet *A* of size '*k*', in which every possible length '*n*' string on *A* occurs exactly once as a substring. Piṅgala demonstrated this for *k* = 2 (*laghu*, *guru*) and *n* = 3 (*gaṇa* of size 3).

**FIGURE 6.2** Binary Cycle of Length 3

SUMMARY

- ▶ The contribution of ancient Indians to the development of mathematical concepts laying the foundation for great scientific discoveries both within India and the rest of the world is significant.
- ▶ A number of archaeological excavations and linguistic evidence provided by the early writers indicate the maturity and supremacy of the Indian mathematical foundations.
- ▶ The Indian numeral system has a long history as is evident from the historical artifacts and archaeological evidence. The origin and the evolution of the numbers could be traced from the time of the Vedic period.
- ▶ The concept of zero was established in India during the period 500–300 BCE and was fully developed by 600 CE.
- ▶ Ancient Indians developed a systematic approach to the number-names that they developed to describe large numbers.
- ▶ The method of using a limited set of numerals but assigning them unique names depending on which place they occupy in the number system is an age-old practice in India.
- ▶ The decimal number system originated in India much before its use in the West.
- ▶ Three generic measures are used in the ancient Indian system pertaining to length, weight, and time.
- ▶ Several measures have been in vogue to measure the length in ancient times in India. Arthaśāstra has elaborate measurements for length suiting different requirements.
- ▶ Time is an important aspect to ancient Indians as they were interested in studying the origin of the Universe. Furthermore, studies pertaining to Jyotiṣa (Astrology and Astronomy) also requires measures pertaining to time.
- ▶ A detailed description of weights is an essential requirement for Āyurveda to mix the ingredients correctly while formulations are prepared.
- ▶ When dealing with mathematical concepts and numbers, poetry is freely used to communicate the same. Unique methods to represent numbers were developed for this purpose.
- ▶ Chandaḥ-sāstra introduces some mathematical concepts including, a formal definition of the word 'śūnya' to denote zero, fundamentals of the binary number system, and some interesting concepts in combinatorial mathematics.

Tarka: The Indian Art of Debate

The debate between great intellectuals is a part of the Indian tradition and it has played a very significant role in advancing knowledge. One of the often-cited famous debates was between Ādi Śaṅkarācārya, the founder of Advaita philosophy and Maṇḍana Miśra, who eventually became his disciple. Maṇḍana Miśra's wife Uabhaya Bhāratī was the moderator of the discussion. The method of debate adopted by the Indians is unique and was highly structured and objective. There was no room for dogma or blind belief in arguments.

The ancient Indian scholars adapted a structured set of argumentative procedures for learning, where using a debate in which two opposing sides presented their views and finally concluded in a mutually acceptable manner by directing the flow of arguments with less deviation from the topic of discussion. The body of knowledge that enabled this is known as Nyāya, which means that by which one is led to a conclusion or a clarified knowledge. This is achieved through discussion or vāda, which is primarily meant for the discernment of the real nature of the thing under investigation and imparting the truth, as one understands it to the other party.

According to Nyāya, a scholastic debate involves two interactive participants—the proponent (one who holds the initial position) and a respondent. The goal of the discussion is the adoption of one of two opposing sides and there is no consideration of victory or defeat. What is adopted is defended using means of right knowledge and its opposite is assailed by confutation, without deviation from the established tenets.

The formal procedure adopted in a vāda can be briefly summarised as follows:

1. The initial proponent puts forward his thesis.
2. If the thesis is thought to be erroneous the respondent may refute it immediately (using step 6 is shown above). On the other hand, if the thesis is accepted, then the respondent asks the proponent to outline the reason for accepting the thesis.
3. The proponent then offers proof outlining the reasons why the thesis should be accepted.
4. The respondent asks if the proof offered is

based on a sound inference and a debate ensues.

5. The proponent negates the faulty relations and erroneous reasoning that may have occurred in the outlining of the proof of his thesis during the debate

This concludes the first part of the debate. If the respondent accepts both the thesis and the proof then the debate concludes. However, if the proof is deemed erroneous for some reason, then the second part of the debate is required.

6. The respondent offers a statement of refutation of the proponent's thesis, which thereby constitutes the initial starting point of his exposition. The refutation that follows aims to demonstrate the errors and inconsistencies of the proponent's position based upon the reasoning and evidence provided by the proponent.
7. The proponent responds with a rejoinder if he thinks that his critic's refutation is in some way erroneous. However, if the proponent accepts the soundness of the refutation, the respondent is asked to state formal proof of the refutation in a positive form.
8. Finally, the respondent offers formal proof of the refutation in inferential form.

Procedural argumentation adopted by Indians tries to resolve knowledge inconsistencies by analysing arguments and providing suitable counter-arguments. Indian logic is based on the conviction that logic is an instrument for the discovery and understanding of reality, which is different from the Western counterpart. Western philosophical systems treat argumentation differently. Western logic does not inquire whether the premises and conclusion of an argument are true. It is solely concerned with determining whether a given argument is valid.

Source: Adopted from Mahalakshmi, G. and Geetha, T.V. (2009). "An Indian logic-based argument representation formalism for knowledge sharing", 17(1), *Logic: Journal of IGPL*, pp. 55–74.

We have made tremendous progress in the last 150 years with technological advances, scientific discoveries, and new inventions. There is a widespread feeling that we have developed capabilities to find new knowledge in hitherto unknown domains, however, this is not a new-found capability for the human race. Human beings armed with the thinking faculty have constantly strived to make innovations. Making such innovations follow a structured process, which culminates in resolving ambiguities in our understanding and finding new knowledge. The entire process requires that we have a robust framework to establish new knowledge. This has been the constant preoccupation of human beings. Ancient Indians too have developed a methodology for establishing new knowledge.

There is a feeling that the Indian thought process is driven by blind beliefs and dogmas rooted in a religious context. It is generally believed that much of the ideas developed and propagated down the ages have been on faith and were left unquestioned. If we explore the culture of developing new knowledge systems in India, we discover something on the contrary. A case in point is the development of Darśanas (Philosophical schools of thought) in India (in Chapter 3 we have introduced Darśanas and discussed some of the unique aspects). There are multiple schools of thought and there has been a healthy tradition of arguing and counter-arguing against the tenets of each school of thought in a highly professional, rational, and objective manner. In fact, one of the Darśanas, known as Nyāya, has specifically focused on how one can systematically inquire into a problem and establish new knowledge in a structured process. Using the tools and techniques developed by Nyāya researchers over 1000 years, a sound framework for logic and epistemology has been developed which has come in handy to critically analyse established tenets and bring in new knowledge. We shall see some aspects of these in this chapter.

7.1 THE KNOWLEDGE TRIANGLE

Knowledge (Jñāna) is defined as apprehension or consciousness. All knowledge is a revelation or manifestation of objects, just as a lamp manifests physical things placed before it. Knowledge may be valid or invalid. Valid knowledge (Pramā) is defined as the right apprehension of an object or knowing an object as it is. A valid knowledge corresponds to the reality and not anything other than that and is indeed produced by some valid means. Invalid knowledge includes memory, doubt, error, and hypothetical reasoning. Memory is not valid because the object remembered is not directly presented to the ātman, but only indirectly recalled. Doubt is uncertainty in cognition. Error is misapprehension as it does not correspond to the real object. Hypothetical reasoning is no real knowledge. When we see a rope as a rope, we have obtained the right knowledge. If we are uncertain whether it is a rope or a snake, we have doubt. We may need some means by which the doubt is unambiguously resolved. If we merely recall the rope that we have seen, it is memory. If we mistake the rope for a snake, we are erroneous.

Valid knowledge is that which has been established rigorously and systematically so that it is non-refutable, true, and reliable in the context and the conditions in which the knowledge is supposed to be applicable. Typically, in the Indian tradition, the aspects of valid knowledge could be best understood from the notion of a knowledge triangle. Figure 7.1 pictorially represents the knowledge triangle, which consists of three components and provides the overall context for obtaining the right knowledge.

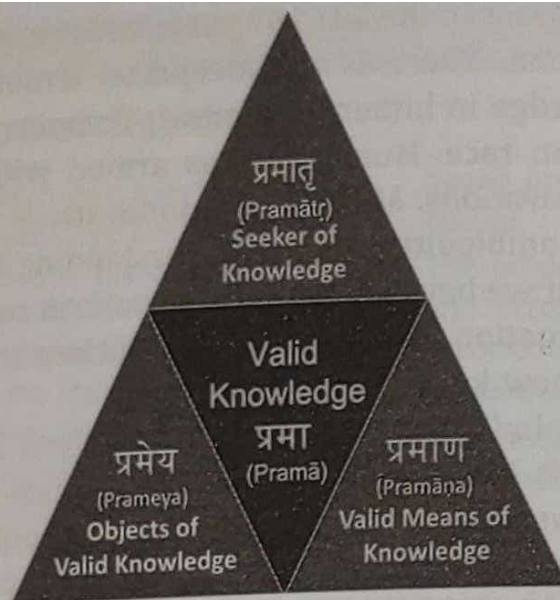


FIGURE 7.1 The Knowledge Triangle

- ◆ Firstly, the knowledge is obtained ultimately only by a seeker of the knowledge. Viewed in this manner all forms of knowledge manifest eventually as 'implicit' knowledge in the mind of the knowledge seeker. Whether it is a Nobel Laureate in Physics or a Scientist working in a laboratory developing a new formulation for medicine, or a spiritual seeker wanting to know who he/she is, the knowledge accrues in him/her only. Therefore, the first component of the knowledge triangle is the seeker of knowledge. The seeker of the knowledge is referred to as **Pramātr**. The knowledge seeker gets mentally involved in the process and commits physical and other resources, and time to the process. Once the knowledge seeker obtain the implicit knowledge he/she may choose to make it available to the others by way of tacit knowledge.
- ◆ The efforts of the knowledge seeker are directed towards some object or entity, as it becomes the context for seeking knowledge. This is the second aspect of knowledge, known as **Prameya**. For instance, a scientist developing a new vaccine will direct his/her entire focus in the knowledge creation process on the virus and all the aspects related to it. A physicist wanting to understand the world as a physical entity will direct the attention toward various aspects of the physical reality. The basic elements that constitute the physical reality, the processes that they employ to combine and regenerate into multiple forms and shapes may become the objects of knowledge that are being sought by the researcher. The term object does not have a mere 'physical' orientation. It includes all that towards which effort is directed to obtain the knowledge, which may include physical, meta-physical, virtual, and other entities.
- ◆ New knowledge creation is invariably a process of starting from known ideas and forms and progressing into the domain of the unknown. In this journey, since the destination is somewhat unknown, we need valid means to assure ourselves that the journey indeed was fruitful, and the process was robust and flawless. Therefore, the third component is to have valid means of obtaining the knowledge. In the Indian tradition, this is known as **Pramāṇa**.

Two schools of Indian thought, Nyāya, and Vaiśeṣika address the issue of knowledge of the reality, i.e. of the physical world of several entities and their interactions. Vaiśeṣika mainly confines itself to 'the exposition of reality' and Nyāya focuses on the issue of 'right knowledge

of reality'. To put it in another way, Vaiśeṣika deals with metaphysics and ontology, and Nyāya deals with logic and epistemology. The key contribution of Nyāya-śāstra lies in providing a robust framework for establishing the right knowledge. The very first sūtra in the text begins with identifying sixteen factors that one needs to take into consideration while establishing the right knowledge. A correct understanding of these and appropriate use of them is critical in the process¹. These sixteen factors provide a comprehensive set of concepts that help one to establish knowledge using a structured approach.

The principles laid out in Nyāya are fundamental and have had the widest appeal as a general framework for creating new knowledge and advancing our thoughts through logic and argumentation. The Nyāya principles have become a very useful tool for experts in all other fields (literature, grammar, philosophical studies, and other fields of knowledge). There have been numerous attempts both in India and outside to write commentaries on Nyāya.

Relating to our knowledge triangle, Vaiśeṣika deals with prameya of the knowledge triangle, whereas Nyāya mainly deals with pramāṇas, valid means of knowledge. However, in Nyāya-śāstra at a generalised level of abstraction 12 objects of knowledge (prameya) have been identified². In this chapter, we shall use the Vaiśeṣika framework to understand the physical reality that presents itself as the object for knowledge creation. On the other hand, we shall use the Nyāya framework to understand various aspects of establishing valid knowledge.